

Moein Razavi

AI/ML Engineer | Data Scientist | Software Engineer

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Professional Summary

Data scientist and software developer with a Ph.D. in Industrial Engineering. Develops AI applications, forecasting and energy decision systems at Ford, with additional experience in wearable health software. Builds data pipelines, machine learning models, cloud APIs, and mobile and web interfaces. Research has 500+ citations on Google Scholar as of September 2026.

Professional Experience

Ford Motor Company

Data Scientist | Aug 2023 - Present

- **Utility data onboarding (HOMeload):** Built a Python/FastAPI pipeline converting customer electricity records into standardized data for energy analysis. Used Vertex AI Gemini for schema discovery, pandas/lxml for extraction, deterministic validation with corrective retries, and BigQuery MERGE for repeatable loads. Deployed with Docker, Cloud Run, and Terraform.
- **EV analytics assistant:** Built a LangChain application using GPT-4o/GPT-4o-mini for intent routing, schema-aware BigQuery text-to-SQL, and retrieval-augmented generation (RAG) with BM25 search over EV manuals and specifications. Added an LLM checker for domain rules and a Dash interface with SAML/ADFS authentication on Cloud Run.
- **Vehicle-to-grid forecasting and dispatch:** Built a Python/PyTorch system connecting vehicle-availability forecasts to fleet selection and energy commitments. Trained TFT/Autoformer models with Optuna on Vertex AI GPUs; combined forecasts with ERCOT prices via gridstatus, Monte Carlo capacity estimates, and greedy vehicle selection. Implemented load reallocation when vehicles disconnect.
- **Rate recommendation:** Developed an application that uses XGBoost electricity-bill predictions, vehicle telemetry, and household consumption to inform utility-plan recommendations. Built Flask APIs, BigQuery workflows, and a Dash interface with SAML authentication; deployed on Cloud Run through Tekton CI/CD.
- **Energy data engineering and reporting:** Built PySpark/BigQuery pipelines and Airflow DAGs for energy-simulator inputs and optimizer outputs. Developed discharge reporting in Looker Studio and REST APIs for operational dashboards.
- **Code analysis and cloud cost estimation:** Built a FastAPI tool using GitPython and Python AST parsing to extract code blocks for LLM-based complexity analysis. Streamed findings through Server-Sent Events; estimated Vertex AI compute costs from model-generated resource assumptions and a static pricing table.

CereVu Medical, Inc.

Software Developer / Research Engineer (Contract) | Jun 2023 - Sep 2024

- **Wearable health platform:** Developed Vitality and Clinical in SwiftUI, using CoreBluetooth BLE and Swift Charts to acquire and display 9 physiological signals. Migrated Clinical from UIKit to MVVM-Cordinator; implemented role-based workflows, calibration, exports, and OAuth 2.0/JWT authentication.
- **Backend and delivery:** Built Python/Flask services with WebSocket biosignal streaming and a React/Next.js dashboard backed by MongoDB. Deployed Docker containers on AWS EC2 with GitHub Actions and distributed iOS builds through TestFlight.

Education

Ph.D., Industrial Engineering | Texas A&M University | Dec 2023

Master of Engineering, Computer Engineering | Texas A&M University | May 2021

Selected Projects

- **Business Action Prioritizer:** Built a Google ADK/Gemini workflow that converts uploaded data and a rubric into ranked actions. Gated execution on plan approval; used PandasAI on local pandas dataframes and an MCP server to publish reports to Cloud Storage.
- **InspecaIytics:** Built SwiftUI and web inspection interfaces with Supabase, AVFoundation photo capture/annotation, and Word/PowerPoint reports. Implemented offline synchronization with SHA-256 change detection and debounced uploads; automated releases with Fastlane/GitHub Actions.
- **Clink Social:** Built SwiftUI messaging and MapKit discovery with Firebase Auth/Firestore/FCM. Implemented feed aggregation and notifications in TypeScript Cloud Functions, plus QR-based cross-device sign-in through time-limited OAuth/JWT token exchange.

Research Experience

Texas A&M University

Research Assistant | Sep 2018 - May 2024

- **Wearable stress detection:** Engineered windowed heart-rate/motion features in Python using data from 54 students over 40 days. Compared classifiers including XGBoost using ROC-AUC and cross-validation; interpreted predictions with SHAP.
- **MERCURY training acceleration:** Co-authored research using random projection with quantization and cached computation reuse; benchmarks showed 1.97x average training speedup across 12 neural networks with accuracy similar to the baseline.
- **OpenSync:** Co-developed an open-source Python platform synchronizing neuroscience sensors and experiment events through Lab Streaming Layer; published benchmarks measured below 0.07 ms RMS recording-time lag.
- **Computer vision:** Developed face-mask and physical-distance monitoring using TensorFlow Faster R-CNN detectors and calibrated perspective transforms to measure distances in video.
- **GlucoseCoach:** Developed an iOS application with a Python/Flask backend for personalized glucose monitoring.

Technical Skills

Languages: Python, SQL, Swift, TypeScript, JavaScript, C++, Java, R, MATLAB

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face, Optuna, OpenCV, time-series forecasting, statistical modeling, deep learning, classification, cross-validation, SHAP/LIME

Generative AI: LangChain, LangGraph, Google ADK, OpenAI Agents SDK, Gemini, RAG, BM25, FAISS, ChromaDB, prompt engineering, text-to-SQL, MCP

Data: pandas, NumPy, SciPy, PySpark, Airflow, ETL/ELT, PostgreSQL, MongoDB, Redis, Firebase, Supabase

Applications: FastAPI, Flask, REST APIs, WebSocket, React, Next.js, SwiftUI, UIKit, CoreBluetooth, MVVM, OAuth 2.0, JWT, SAML

Cloud: AWS (S3, EC2, Lambda, SageMaker), GCP (Vertex AI, BigQuery, Cloud Run, Dataproc), Azure OpenAI

Delivery: Git, Docker, Kubernetes, Terraform, CI/CD, Tekton, GitHub Actions, AWS CodeDeploy, Fastlane, TestFlight, XCTest

Selected Publications

Razavi, M., et al. Machine Learning, Deep Learning, and Data Preprocessing Techniques for Detecting, Predicting, and Monitoring Stress and Stress-Related Mental Disorders: Scoping Review. JMIR Mental Health, 2024.

Janfaza, V., Weston, K., Razavi, M., et al. MERCURY: Accelerating DNN Training By Exploiting Input Similarity. IEEE HPCA, 2023.

Razavi, M., et al. OpenSync: An open-source platform for synchronizing multiple measures in neuroscience experiments. Journal of Neuroscience Methods, 2022.

Razavi, M., et al. An Automatic System to Monitor the Physical Distance and Face Mask Wearing of Construction Workers in COVID-19 Pandemic. SN Computer Science, 2022.